

Desiree Fisker

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EDUCATION

McGill University

Doctor of Philosophy in Mechanical Engineering

Dynamics, Estimation, and Control in Aerospace and Robotics Lab, led by Prof. James Forbes

September 2025 – Present

University of Toronto

Master of Applied Science in Aerospace Engineering

Collaborative Specialization in Robotics

Autonomous Space Robotics Lab, led by Prof. Tim Barfoot

September 2023 – August 2025

- CGPA: 3.62
- **Notable Courses:** State Estimation, Robot Motion Planning, Perception for Robotics, UAV Design
- **Teaching:** *First-Year Calculus TA Coordinator*
Collaborated on course structure, aided in implementing new fundamental concept review sessions for 1100 students, organized grading and hours for 22 TAs, and ensured cohesive tutorial content

Queen's University

Bachelor of Applied Science in Engineering Physics

Mechanical Sub-Specialization

16-Month Professional Internship

September 2018 – April 2023

- CGPA: 3.55, Last 2 Years' GPA: 4.00
- **Notable Courses:** Mechanics, Introduction to Robotics, Automatic Control, Thesis
- **Teaching:** *Engineering Design 200A/200B TA*
Graded reports and presentations based on a rubric and valued industry experience, provided supplemental instruction on technical course materials, and managed four teams of 5 students

PUBLICATIONS AND MANUSCRIPTS

- [1] **D. Fisker**, A. Krawciw, S. Lilge, M. Greeff, and T. D. Barfoot, "UAV See, UGV Do: Aerial Imagery and Virtual Teach Enabling Zero-Shot Ground Vehicle Repeat," *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 17518–17525. [IEEE Xplore: 10.1109/IROS60139.2025.11247387](https://doi.org/10.1109/IROS60139.2025.11247387)
- [2] **D. Fisker**, "Zero-Shot Ground Vehicle Navigation Enabled by Aerial-Based Digital Twin Scene Reconstructions," M.A.S. thesis, Department of Aerospace Science and Engineering, University of Toronto, 2025. [University of Toronto Scholaris](https://utoronto.scholaris.ca/handle/1807/117444)
- [3] A. Zhong, B. Beggs, C. Vilorio, **D. Fisker**, J. Fisman-Guarascio, T. Lahteenmaa-Swerdlyk, "Pneumatic Model Plane Launching System", Department of Physics, Engineering Physics and Astronomy: PHYS/ENPH 454 Advanced Engineering Physics Design Project Collection, 2023 <http://hdl.handle.net/1974/31460>

SKILLS AND ABILITIES

Programming Languages:

Python (PyTorch, PySTEAM, Pylgmach, NEAT, TensorFlow), C++, Arduino, MATLAB

Software Competencies:

SolidWorks (FEA, Large Assemblies, Drawings), ANSYS (FEA), Git, LTSpice, KiCAD (circuitry design, PCB manufacturing), LaTeX, VSCode, Ubuntu, Docker, ROS 1, ROS 2, Rviz, Gazebo

Design and Manufacturing:

Drill Presses, Lathes, Mills, Taps, Saws/Blades, Soldering, 3D Printing (slicing, PLA/ABS/Resin, infill variation), Oscilloscopes, Multimeters

Testing and Mobile Platforms:

Wind Tunnel, Carbon Fiber Filament Winder, Hydrostatic Pressure Chamber, Explosives Bunker, Small-Scale UAV/UGVs (Pixhawk, Arduino)

RESEARCH AND INTERESTS

Robotics, Systems Engineering, UAVs/UGVs, Aerospace Engineering, Guidance and Navigation, Perception

- **Virtual VT&R via NeRF World Representations** [🔗]

Tools: NeRFStudios, Instant-NGP, Colmap, Visual Teach and Repeat (VT&R3), ROS 1/2, Gazebo

Goal: Enable autonomous path following for a UGV within a mapped environment by modeling it with NeRFs trained on UAV imagery and extracting features from novel viewpoints.

- Investigated and tuned optimal parameters for training a NeRF scene representation to produce clear and differentiable novel viewpoints of a large-scale scene.
 - Compared physical and virtual sensor models to extract and align depth and feature information for dataset construction.
 - Developed a pipeline that allows virtual piloting of a UGV in a simulation constructed from a drone image-trained NeRF.
- **Autonomous UAV Navigation**
- Tools:** Google Earth Studios, VICON, Instant-NGP, Colmap, Pi-NeRF
- Goal:** Autonomously localize a UAV below 30m AGL, without GNSS, using Google Earth satellite imagery and Inverse NeRFs.
- Adapted an invertible neural network and Mip-NeRF implementation to produce pose estimations of novel query images within 5° and 0.05 units 90% of the time.
 - Created image-pose pair datasets by tracking multiple reference frames using a stereo camera in a VICON lab.
 - Evaluated Google Earth 3D renders versus real-life images using established feature matching techniques (SIFT, SURF, ORB, SuperPoint).
- **RRT Path Planning and Deep Reinforcement Learning**
- Tools:** RRT, RRT*, DRL, DeRRT*, RNN, HMM
- Goal:** Generate an improved initial solution from RRT by leveraging previous sampling experiences through DRL.
- Modified the RRT sampling algorithm to incorporate a sequence model that conditions future states on past states.
 - Developed a baseline system for learned experiences using Hidden Markov Models and RNNs.
- **Trajectory Optimization for Solar-Powered UAVs with NEAT**
- Tools:** PyTorch, PySolar, Neural Evolution of Augmenting Topologies Package, SolarNEAT
- Goal:** Optimize the trajectory of a solar-powered UAV to maximize flight time on a single battery charge.
- Tuned the NEAT genetic algorithm to generate effective neural network architectures for SolarNEAT.
 - Constructed Python simulations to validate and ensure feasible flight dynamics.
 - Integrated Receding Horizon Control into SolarNEAT, reducing computational cost.
 - Achieved sustained solar-powered flight for 28.5 hours by taking advantage of the day-night cycle to recharge the battery.

CERTIFICATES, HONOURS, AND AWARDS

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| • FRQNT Doctoral Research Scholarship:
\$25,000/year for four years (\$100,000 total) | • Queen's University Principal's Scholarship: \$7,000 |
| • NSERC USRA: \$6,000 for research at the Royal Military College of Canada | • MITx Certificate of Achievement
(Massachusetts Institute of Technology)
September 2016 – December 2016 |
| • Queen's University Dean's Scholar: First Class Honours Graduate | ◦ Introduction to Aerospace Engineering: Astronautics and Human Space Flight |
| | ◦ GPA: 4.00 |

PROFESSIONAL EXPERIENCE

- **Cesaroni Technology Inc.**  May 2021 - August 2022
Aerospace Engineering Intern
- Contributed to large-scale testing facility CAD assemblies to ensure ergonomic manufacturing and use.
 - Collaborated on a power budget for a thrust stand DAQ system and designed and completed a mass budget for a thrust stand preload system to reduce load cell noise.
 - Assisted designing and independently implemented and tested new methods for signal monitoring and data acquisition in hazardous environments.
 - Optimized shear pin locations on retaining ring and airframe components using mathematical models and FEA on Solidworks CAD models.

- Designed and integrated load cell mounting system for large-scale thrust stand that applied known force to initiate DAQ and dampen background noise.
- Improved efficiency and error margins for the production and testing of rocket motors by creating documentation.
- **Royal Military College of Canada Advanced Aircraft Design Lab** [🌐] *May 2020 - December 2020*
Aerospace Engineering Research Assistant
 - Fixed PID tuning for open source/custom guidance system by troubleshooting field testing results with precise navigational commands to improve autonomous data collection for mapping.
 - Integrated new hardware and software for vehicle attitude and stabilization control on UAVs and UGVs.
 - Advanced timeline by creating MATLAB and Python programs to assess the accuracy of waypoint and obstacle determination for path planning by compiling collected magnetic field data into topographical-style maps.
 - Independently conducted outdoor navigation testing on autonomously determined waypoints based on gathered data for the local area by creating required manoeuvres and UGV to return to home requests.
 - Created SOP and start-up documentation for magnetic field-based autonomous guidance system.

VOLUNTEER EXPERIENCE

Queen's Rocket Engineering Team [🌐]

- **Chief Operations Officer** *July 2022 - June 2023*
 - Worked closely with Queen's faculty to pioneer new alumni funding frameworks for design teams by creating strategic plan that was presented to the Dean of Engineering.
 - Provided guidance based on experience to subteam leads on how to host meetings, efficiently conduct effectively research, and engage team members whilst ensuring a sound design concept is developed.
 - Oversaw report writing and deadlines to ensure relevant information was presented correctly and on time.
- **Chief Technical Officer** *July 2020 - June 2021*
 - Optimized performance and durability of a sounding rocket by using knowledge of rocket aerodynamics to evaluate design choices for composite, mechanical, and electrical parts.
 - Significantly enhanced rocket subsystem cohesiveness and team interactions by using position as CTO to act as a liaison between subteams and help get more members into multidisciplinary roles.
 - Established vibration and ejection testing, subsystem continuity check methods, and more accurate payload testing scenarios which ensured design functionality.
- **Payload Team Lead** *July 2019 - June 2020*
 - Improved previous team operations by delegating tasks to team members, managing overall progress of project from a high level while completing deliverables, and teaching required technical skills.
 - Developed novel 'Double Split' parachute deployment mechanism as a purely mechanical system to reduce complexity and improve functionality by 50%.
 - Condensed technical research, documentation, and testing results to fulfil requirements of competition reports and to be understood by judges who do not have extensive background in the given topics.
 - **Project: Cryogenic Dark matter Detection Proof of Concept Experiment**
 - * Researched and designed a payload similar to DAMIC from SNOlab, that used CMOS sensors to detect WIMPs via tracking anisotropic trajectory distributions, benefiting from lack of interference from earth.
 - * Conducted testing of all subsystems including cryogenic cooling, physical data acquisition, event sequencing, and fixed pattern noise, suppressing post-processing.
 - **Project: Exo-Planetary Drone**
 - * Developed concept and led R&D for an exo-planetary exploration drone that can be deployed and ride thermal currents to collect relevant atmospheric data.
 - * Specified and integrated sensors on SRAD DAQ PCB to collect data, physically tested control feedback loop based on mathematical trajectory modelling from MATLAB, and created power distribution schematic.
 - * Led team members through the creation of a mathematical model for the predetermined trajectory of the drone from deployment to landing.

ADDITIONAL EXPERIENCE

- **Skydive Toronto Inc. [🌐] and Parachute Montreal [🌐]** *June 2021 - Present*
Instructor, Coach, Video Flyer, and Packer
 - Provided adaptive teaching methods to novice skydivers to suit their learning style so manoeuvres could be safely learned and executed.
 - Instructed life-saving emergency procedures to ensure students can effectively conduct them in-air.
 - Conducted and taught all post-gear up, pre-boarding, pre-jump safety continuity and gear checks.
 - Facilitated the learning of best practices and efficient main canopy packing skills to novice skydivers.
 - Doubled tandem packing capacity, increasing daily operations to provide over 100 tandems/day.
 - Packed over 700 Sigma and Micro Sigma tandem parachutes and 800 Sport parachutes with no malfunctions and clean openings.

PROFESSIONAL MEMBERSHIPS

- **IEEE, Student Member**, Membership ID: 100800206 *September 2023 - Present*
- **Scuba Schools International**, Membership ID: 3711454 *June 2023 - Present*
- **Coaching Association of Canada**, Membership ID: 6578295 *June 2022 - Present*
- **Canadian Sport Parachuting Association**, Membership ID: 23609 *June 2021 - Present*

HOBBIES AND ACCOMPLISHMENTS

- Sport Skydiving: 900+ Solo Skydives made over six summer seasons
- Scuba Diving: NAUI Rescue Scuba Diver, Advanced Open Water Certified, 12 Cenote Cavern Dives
- Rock Climbing: Bouldering, Top Rope Certified
- Notable Backcountry Camping Experience: three week-long ocean kayak camping trips and one three-day backcountry winter camping trip
- Wood Working: Folding Muskoka Chair, Magnetic Levitating Fan

REFERENCES

1. **Timothy D. Barfoot**
Professor, Autonomous Space Robotics Lab
University of Toronto Institute for Aerospace Studies, University of Toronto
Email: Upon Request
Phone: Upon Request
Relationship: MASc Thesis Supervisor
2. **Melissa Greeff**
Assistant Professor, Robora Lab
Electrical and Computer Engineering, Queen's University
Email: Upon Request
Phone: Upon Request
Relationship: MASc Thesis Advisor
3. **Bryan Pilon**
Aerospace Engineering Director
Cesaroni Technologies Inc., Aerospace Department
Email: Upon Request
Phone: Upon Request
Relationship: Supervisor
4. **Ruben Perez**
Professor, Advanced Aircraft Design Lab
Royal Military College of Canada
Email: Upon Request
Phone: Upon Request
Relationship: Research Assistant Supervisor and Undergraduate Thesis Supervisor
5. **Jordan Morelli**
Professor
Queen's University
Email: Upon Request
Phone: Upon Request
Relationship: Capstone Project Supervisor